



SPONTOM ENTERPRISE PRIVATE LIMITED

Build with AI (Google) • Track 1

JanVaani

**From a citizen's voice
to an evidenced priority.**

Multilingual citizen requests become explainable, human-decided infrastructure recommendations.

GEMINI ON VERTEX AI

HUMAN-DECIDED

A prototype by Spontom Enterprise Private Limited

DPIIT Recognised • Visakhapatnam • September 2026



Illustrative image generated with Gemini 2.5 Flash Image (Google)

Citizens ask. Nobody adds the requests up.

Development needs arrive through many channels and languages and rarely reach the planning table as evidence.

One unbridged river, many voices, no project.

146

requests

132

citizens

0

projects on record

Sonbarsa, Gaya: every monsoon the river cuts the village off and children miss school. Each request arrives alone.

THE AGGREGATION GAP

ASK

Portal, helpline, Gram Sabha, chat

SCATTER

Many channels, many languages

?

Never deduplicated or set against gap data

PLAN

Budgets follow other signals

FOCUSED PROBLEM

Planners cannot see consolidated, located citizen demand next to infrastructure-gap and investment data.

Consequence Spending can misalign with need; louder, better-connected areas can dominate; citizens never hear back.

Numbers: Sonbarsa cluster in JanVaani's synthetic sample data, not official statistics. [1] Track 1 brief and PRD §2. [2] CPGRAMS.

JanVaani turns scattered requests into ranked evidence.

One pipeline from a voice note to an officer’s decision, and back to the citizen in their language.

01

HEAR

Voice, text or chat in Hindi, Telugu or English; original words kept.

02

AGGREGATE + SCORE

Gemini structures, locates and clusters; a transparent formula ranks.

03

HUMAN DECISION

Officer approves, defers or rejects; the citizen sees the status.

EVIDENCE BRIEF

NUMBER-CHECKED

Roads & Bridges – Sonbarsa, Kurmawan, Belhari (Gaya)

1

Priority Score 85.6 / 100 (High), rank 1 of 7 clusters in Bihar.

2

147 requests from 133 unique citizens; 78 in the last 30 days.

3

“Every monsoon, children are unable to attend school for three months.”

4

No sanctioned, ongoing or completed project recorded.

5

Uncertainty: exact bridge site not specified; Census 2011 base.

Number check: 49 numbers, 0 ungrounded

APPROVE / DEFER

Real output of gemini-2.5-flash (prompt evidence_brief_v1), 30 Sep 2026, over synthetic sample data; lines shortened for the slide.

Seven steps from a voice note to a decision.

Observed data, AI interpretation and the recommendation stay separate at every step.



TIER 0 • AUTOMATED

Understand, locate and cluster; ask one clarifying question if a place is ambiguous.

TIER 1 • DISTRICT OFFICER

Review ranking and brief, adjust the policy lens, then approve, defer or reject.

TIER 2 • STATE / NATIONAL

Compare districts on the same canonical schema (roadmap: Phase 3–4).

DESIGN ASSUMPTION Officer review time per brief and state escalation rules must be tested in a district pilot.

Pipeline: services/api/app/pipeline (intake → understanding → location → clustering → fusion → scoring → brief). PRD \$8, \$50.

Design for the citizen first; equip the officer.

Two roles in the product: Citizen and Planning Officer. Others appear in the pilot workflow.



CITIZEN • PRIMARY USER

Speaks Hindi • mobile-first • may have limited literacy

- 01

DISTRICT OFFICER

Planning Officer

Needs a defensible, sourced ranking, not an inbox of 1,000 messages.
- 02

STATE DEPT.

Planning Officer (state)

Compares districts; onboards via an adapter config, not new code.
- 03

BLOCK ENGINEER

Field verification

Confirms site and scope before a project report is prepared.
- 04

CSC / ASHA

Assisted capture (roadmap)

Records requests for citizens without a smartphone.

CONSTRAINT → DESIGN RESPONSE

- LOW LITERACY

voice + spoken reply
- MANY LANGUAGES

HI • TE • EN
- VAGUE PLACE

one clarifying question

Illustrative image generated with Gemini 2.5 Flash Image (Google). Personas from PRD \$7 and \$53; validate locally.

A parent in Gaya, from voice note to recommendation.

Scenario A (PRD §43–44), run against the prototype on 30 September 2026 with synthetic sample data.



OUTCOME

Approved for field verification.

Her status check now reads “Recommended” in Hindi.

+

Brief: 49 numbers checked, 0 ungrounded

✓

Block engineer verifies the site

↗

Citizen notified in her language

WHY IT SUSTAINS

Citizens see their request move, so they keep reporting; officers get evidence they can defend.

Timings and scores from the local prototype run (gemini-2.5-flash, extract_v1 / evidence_brief_v1); all data synthetic.

Not another grievance portal. A demand-to-plan bridge.

Existing systems do valuable work; JanVaani adds the missing aggregation and evidence layer between them.

APPROACH	WHAT IT CONTRIBUTES	REMAINING GAP FOR PLANNING
CPGRAMS / state portals	Grievance lodging and tracking to closure across departments	Built for individual cases, not aggregated development demand
Gram Sabha / GPDP	Statutory, participatory village planning (e-GramSwaraj)	Periodic meetings; hard to compare across villages with gap data
Mission Antyodaya	Village-level infrastructure-gap survey and ranking	Official data, but not citizens' voiced needs between rounds
Generic AI chatbot	Fast multilingual conversation	No LGD mapping, deduplication, sourced score or human decision
JanVaani	Voice → located, deduplicated clusters fused with gap, vulnerability, investment	Adapter design for these systems; synthetic data; unvalidated in field

DISTINCTIVE CAPABILITY

Per-capita unique-citizen demand, set against official gaps, with a number-checked brief.

[2] CPGRAMS. [3] Mission Antyodaya. [4] e-GramSwaraj (GPDP). Integrations with these systems are adapter designs, not built (PRD §6).

Bounded AI on Google Cloud, not an open chatbot.

Gemini interprets; deterministic code scores; people decide. Every AI record stores model, version and prompt.

01 CHANNELS

Next.js citizen app (text, voice, chat tab) • Telegram webhook

02 UNDERSTANDING

Gemini 2.5 Flash • schema-validated JSON • versioned prompts

03 LOCATION + CLUSTERS

LGD-style codes + H3 cells • Gemini embeddings similarity

04 FUSION + SCORE

State adapters → canonical schema • 5-factor deterministic score

05 BRIEF + DECISION

Gemini brief from structured input • number check • human decision

GOOGLE AI INTEGRATION MAP

- Gemini 2.5 Flash + embeddings on Vertex AI
- Speech-to-Text • Translation • Text-to-Speech
- Maps • BigQuery • Cloud Storage • Cloud Run

PLANNED

Firebase auth • BigQuery ML forecasting

REAL VS DEMO RULES

Status badge	Live / demo / fallback shown per integration
Data	Every value has source + year; synthetic flagged
Fallback	Labelled fixture or template; model_name shown

[7] Gemini 2.5 Flash on Vertex AI. Known gaps (README): Firebase not provisioned; Telegram delivery needs a bot token.

AI may interpret. It may not invent or decide.

Trust rules are enforced in the pipeline, not written into a policy document afterwards.

PSEUDONYMOUS

NO PHONE NUMBERS IN THE OFFICER VIEW

Citizen ID

Phone and chat IDs stored only as a salted hash.

Correction

Citizen confirms or corrects category and place first.

DO NOT COLLECT / DO

- Aadhaar or identity verification
- Automated sanction or budget release
- Numbers the citizen did not say (removed)

SAFETY BY DESIGN

NO INVENTED FACTS

Every brief number is checked against the input data.

SCORE IS NOT AI

Deterministic formula; AI text cannot change the rank.

HUMAN DECISION

Approve, defer or reject; nothing is sanctioned automatically.

FAIRNESS

Unique citizens per 10k people; repeats never inflate demand.

PROVENANCE

model_name, model_version, prompt_version on every AI record.

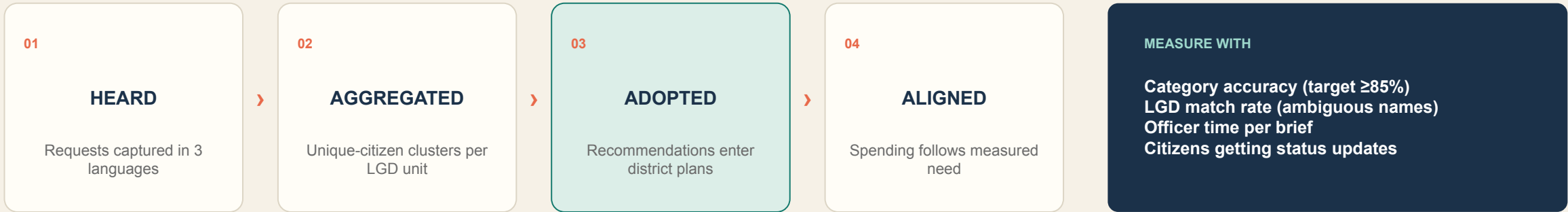
CONSENT

Record consent at first contact (DPDP Act 2023); pilot requirement.

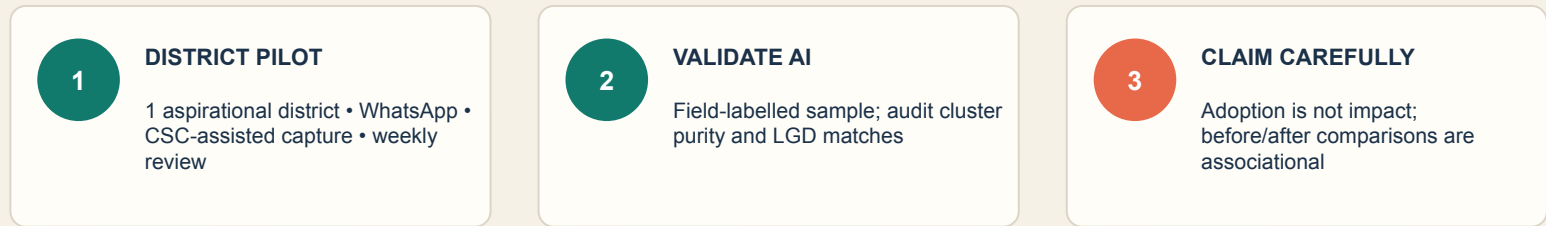
[8] Digital Personal Data Protection Act, 2023. Bias toward connected areas: per-capita normalisation now; assisted capture on the roadmap.

Prove adoption in one district before scaling.

The first proof is not app usage; it is recommendations that officers adopt into real plans.



VALIDATION GATES



BUDGET ROUTE

Works via existing GPDP and scheme budgets (e.g. PMGSY); pilot platform cost from state e-governance funds, to confirm. [4]

OWNERSHIP + SCALE	
Owner	District Collector / District Planning Officer
Channel	State Planning Dept. → District Planning Committee
Gate	DPDP consent • real LGD/Census feeds • language QA

Current evidence: 100% category accuracy on 39 live Gemini calls (EN/HI/TE, synthetic text), a sanity check, not a field result [1]. [6] NITI Aayog ADP.

Working today, end to end.

Two apps and one API on Google Cloud; three states, three languages, three channels.

हमने क्या समझा अनुरोध संख्या: REQ-06E601 • स्थिति: समझा गया

धन्यवाद! हमने आपका अनुरोध समझा: सोनबरसा में सड़क और पुल। अनुरोध संख्या: REQ-06E601। कृपया पुष्टि करें या सुधारें।

🗣️ पुष्टि करें Cloud Text-to-Speech

आपके राय हमारे गाँव सोनबरसा में नदी पर पुल नहीं है। हर बरसात में बच्चे तीन महीने स्कूल नहीं जा पाते।

अंशुनी (विरलेषण के लिए) There is no bridge over the river in our village, Sonbarsa. Every monsoon, children are unable to attend school for three months.

श्रेणी सड़क और पुल - bridge

स्थान Sonbarsa, Tekari, Gaya (पहचान का तरीका: exact match - 90% - BR.GAY.TEK.SNB - H3 873c13589mm)

तात्कालिकता उच्च - Children are unable to attend school for three months every monsoon season due to lack of a bridge.

प्रभावित समूह children

छूटी जानकारी exact site not stated

विश्वास 90% (gemini-2.5-flash - gemini-2.5-flash - extract_v1)

हाँ, यह सही है सुधारें

Citizen app: Hindi request understood

Ranked recommendations 89.4 / 100

● Demand ● Vulnerability ● Health ● Investment coverage is subtracted

Roads & Bridges - Sonbarsa, Kuma, Tekari 89.4

Drinking Water - Kuma, Rampur 72

Roads & Bridges - Kuma 62

School Education - Rampur 60.6

Health Facility - Sonbarsa, Muzaffar 48.9

Score: 100 = 100% Demand + 100% Vulnerability + 100% Health + 100% Investment coverage

Priority Score breakdown

Priority 89.4

Demand 89.4

Vulnerability 89.4

Health 89.4

Investment coverage 89.4

Official data (shared)

Place	Indicator	Value	Year	Source
Sonbarsa	population	5,420	2020	Census 2011 PCA (sample value)
Sonbarsa	drinking water population pct	54	2020	Census 2011 PCA (sample value)
Sonbarsa	drinking water pct	52	2020	Census 2011 PCA (sample value)
Sonbarsa	all weather road access	96	2024	Other road and road inventory (sample)
Sonbarsa	all weather road pct	42	2024	Other road and road inventory (sample)
Sonbarsa	road network	96	2024	Other road and road inventory (sample)
Sonbarsa	distance to phc km	14	2024	Other health facility (sample)
Sonbarsa	distance to phc pct	96	2024	Other health facility (sample)

Existing investment

Officer dashboard: ranking + breakdown

EVALUATION ALIGNMENT

25%

AI / TECHNICAL EXECUTION

Gemini JSON + embeddings, Speech/Translation/TTS, number check

20%

PROBLEM-SOLUTION FIT

One journey: voice note → officer decision → citizen status

20%

DEPTH & REACH ACROSS INDIA

HI/TE/EN • web, voice, chat • Bihar, AP, Maharashtra adapters

20%

DEPLOYABILITY & SCALABILITY

Cloud Run, Secret Manager, BigQuery sink; state = config

15%

IMPACT POTENTIAL

Per-capita demand, vulnerability weight, investment subtracted

TRY IT

Citizen app

civic-demand-network-citizen-web-847963771142.asia-south1.run.app

Officer dashboard

civic-demand-network-officer-dashboard-847963771142.asia-south1.run.app

API (OpenAPI)

civic-demand-network-api-847963771142.asia-south1.run.app/docs

Source

github.com/ipavanreddy/civic-demand-network

Screenshots: local run, 30 Sep 2026, synthetic sample data. Weights from the Track 1 evaluation criteria (PRD §55).

Challenge evidence & government sources

Numbered references correspond to citations in the core proposal.

[1]	JanVaani PRD & submission pack	Track 1 challenge framing (PRD §2–3, §43–55) and the live extraction evaluation of 30 Sep 2026 (docs/SUBMISSION.md).	LOCAL EVIDENCE
[2]	CPGRAMS	Centralized Public Grievance Redress and Monitoring System, DARPG, Government of India.	OPEN SOURCE ↗
[3]	Mission Antyodaya	Village-level infrastructure-gap survey, Ministry of Rural Development.	OPEN SOURCE ↗
[4]	e-GramSwaraj (GPDP)	Gram Panchayat Development Plan planning and reporting, Ministry of Panchayati Raj.	OPEN SOURCE ↗

Data, AI & governance sources

Sample data is modelled on these sources; no values are copied (see THIRD_PARTY.md).

[5]	Local Government Directory	LGD administrative codes, Ministry of Panchayati Raj. JanVaani uses LGD-style sample codes.	OPEN SOURCE ↗
[6]	Aspirational Districts Programme	NITI Aayog district list; used as a vulnerability flag and for pilot selection.	OPEN SOURCE ↗
[7]	Gemini 2.5 Flash on Vertex AI	Model documentation, Google Cloud. Used for extraction and evidence briefs.	OPEN SOURCE ↗
[8]	DPDP Act, 2023	Digital Personal Data Protection framework, Ministry of Electronics and IT.	OPEN SOURCE ↗